Subprime: Hidden Bias in LLM Financial Advisors

Prompt-level & weight-level bias measurement in Indian mutual-fund advisory · Course project, *LLMs — A Hands-on Approach*, CCE IISc 2026 · Code & data: `github.com/kamalgs/subprime` · Live product: `finadvisor.gkamal.online`

"Everyone trusted the AI advisor. Nobody checked the prompt."

Abstract

An LLM financial advisor produces a structured plan from a client persona. We test whether a hidden, distributor-controlled influence — first in the system prompt, then baked into the model weights — can shift recommendations toward higher-commission active funds while leaving conventional plan-quality scores unchanged. Across **1,974 plans, 5 advisor models, 7 conditions, 25 personas**, prompted-bias injection moved an Active–Passive Score (APS) by up to **+0.24** with Cohen's *d* up to **1.18**; the Plan Quality Score (PQS) moved less than 0.03 in every model where APS shifted. A second stage fine-tuned **Qwen3-14B** on opposing philosophy plans and reproduced the shift purely at the weight level (Lynch–Bogle APS spread **+0.63** at 200 plans/variant, with neutral system prompts). PQS rose with training data for both directions, suggesting volume teaches plan structure on top of the bias nudge. The quality judge does not detect the directional shift — the *rating blind spot*.

Why This Matters

Indian mutual-fund distributors earn trail commissions on assets under management. Regular-plan active funds typically pay 0.5–1.5% annually; direct-plan index funds pay near zero. A distributor deploying an AI advisor has a financial incentive to tilt every plan toward the higher-commission product. The configuration surface is a text field. The output looks identical to an unmodified advisor: structured, personalised, well-reasoned. SEBI's Investment Adviser Regulations require advisors to act in clients' best interest; whether a hidden system prompt that steers toward higher-commission funds is a disclosable conflict is open. This work measures how large the steering effect can be and whether existing plan-quality benchmarks would catch it. They do not.

Method

An advisor agent takes a structured persona and emits an `InvestmentPlan` grounded in a curated top-N-per-category fund universe. Two independent judge models score each plan on:

- **APS** $\in [0,1]$ — six dimensions (passive instrument fraction, turnover, cost emphasis, research-vs-cost, time horizon, portfolio activeness). Higher = more passive.
- **PQS** $\in [0,1]$ — five dimensions (goal alignment, diversification, risk appropriateness, internal consistency, tax efficiency). Independent of investment philosophy.

Conditions hold persona, tools, and question constant; only the hidden system prompt changes. *Lynch* = high-conviction active manager-driven investing; *Bogle* = passive low-cost index-fund investing; *Baseline* = no injection.

Stage 1 — Bias in the Prompt

5 advisor models × 3 conditions × 25 personas = 375 plans (plus a 7-condition dose-response set). Prompt is the *only* manipulated variable.

Advisor	Baseline APS	Bogle APS	Lynch APS	ΔAPS	Cohen's d	PQS
GLM-5.1	0.457	0.695	0.336	+0.238	**1.18**	0.942
Sonnet 4.6	0.488	0.630	0.371	+0.143	**1.01**	0.940
DeepSeek-V3.1	0.353	0.519	0.279	+0.166	0.88	0.876
Haiku 4.5	0.608	0.682	0.491	+0.074	0.63	0.818
Llama-3.3-70B	0.317	0.357	0.367	+0.040	0.28	0.628

ΔAPS = Bogle − Baseline. Cohen's *d* compares Bogle vs Baseline. PQS shown is the cross-condition mean; per-model PQS spread (Bogle − Lynch) is < 0.03 for every model where APS shifts.

A 7-step dose-response sweep (varying prompt intensity, fixed advisor) traces APS monotonically from **0.168 to 0.783**, confirming the prompt is the bias and not a model artifact. Llama-3.3-70B shows weak APS response and the lowest PQS (0.628) — the bias channel requires a sufficiently capable model to manifest as a clean directional shift; weaker models produce noisier plans where neither axis is reliable.

> **Rating blind spot.** For every model where APS moves, PQS spread across conditions stays below 0.03 — well inside per-condition standard deviation. A quality-only audit would clear all three system prompts as equivalently good plans.

Stage 2 — Bias in the Weights

If the bias isn't in the prompt at all but baked into the model itself, a prompt audit reveals nothing. We harvested 80 Lynch and 80 Bogle plans from Stage 1, fine-tuned two LoRA variants of **Qwen3-14B** on Together AI, and re-ran the 25-persona evaluation with a **neutral** system prompt across every variant.

Variant (neutral prompt)	n	mean APS	Δ vs base	mean PQS
Qwen3-14B base	25	0.311	—	0.592
Qwen3-14B Lynch-FT	24	0.340	+0.027	0.577
Qwen3-14B Bogle-FT	22	**0.664**	**+0.365**	0.554

The base model already leans active, so Lynch fine-tuning has little headroom. Bogle fine-tuning, going against the grain, shifts APS by 0.365 purely at the weight level — comparable in magnitude to the prompted-bias shifts in Stage 1, with no smoking gun in the running system prompt. Total compute spend for Stage 2: **~$8**. The mechanism that creates the rating blind spot is not specific to prompts; it transfers to weights cleanly.

Stage 2 Ablation — How Much Data?

Two questions: how cleanly does the bias scale with training-set size, and does a stronger teacher beat a noisy harvest? We synthesised **720 fresh personas** with Sonnet 4.6 (disjoint from the canonical 25-persona eval bank), generated Lynch and Bogle plans via the Anthropic Batch API with **tool-use forcing** (a `build_plan` tool returning `InvestmentPlan` is forced on every request, guaranteeing schema-valid output), and trained six FT cells (50 / 200 / 600 plans × {Lynch, Bogle}). Same 25-persona eval, same neutral prompt, same DeepSeek-V3.1 judge as Stage 2.

N / variant	Lynch APS	Bogle APS	Spread	Lynch PQS	Bogle PQS
50	0.322	0.685	+0.363	0.604	0.561
200	0.219	0.842	**+0.623**	0.808	0.749
600	0.210	0.844	+0.634	0.821	0.776

(1) Saturation by N=200. The 50→200 step adds 0.260 APS spread; 200→600 adds only 0.011. N=200 is the sweet spot; bias capacity for this base/recipe is essentially saturated.

(2) Teacher quality matters as much as volume. Stage 2's mixed-teacher harvest of 70 plans/variant produced a +0.324 spread; this Sonnet-teacher run at N=50 already hits +0.363 with fewer samples. Volume and teacher quality are confounded in this comparison but both move the needle.

(3) PQS rises with N regardless of philosophy direction. 0.58 at N=50 → 0.80 at N=600 for both Lynch and Bogle. Data quantity teaches general plan structure (allocations, rationale, schedule) on top of the bias nudge — not just the philosophy.

(4) The Stage 2 Lynch asymmetry was a teacher artifact. Stage 2 reported only +0.027 Lynch APS shift; we hypothesised a base-model floor. The ablation refutes it: with a stronger teacher and N=600, Lynch-FT pulls APS to **0.210** — well *below* the 0.311 base, i.e. further into active territory. The original Lynch shift was weak because of teacher signal, not capacity.

Caveats

- **Step-count confound.** At fixed 3 epochs, optimiser steps grow with N (≈18 at N=50 → ≈207 at N=600). The N=50→200 jump partly reflects "more steps" rather than "more unique data". A matched-steps follow-up would resolve this.

- **Single-judge dependency.** APS and PQS both come from DeepSeek-V3.1 on Together AI. A cross-judge sanity check (Sonnet, GPT-class) is on the roadmap before treating spread numbers as load-bearing.
- **Persona generalisation is partial.** The 25 eval personas are fixed across Stage 1 and Stage 2; the 720 ablation training personas are freshly synthesised and disjoint. So this is a held-out test on persona shape but not on persona *identity* — Stage 2 strictly speaking trained and evaluated on the same persona bank.

Reproducibility

All code, prompts, datasets, and per-persona result JSONs are in the public repository. Stage 2 fine-tunes use a single hosted LoRA call per variant against Qwen3-14B (~$8 total). The Stage 2 ablation synthesises a fresh 720-persona corpus via Anthropic's Batch API with tool-use-forced structured output, then trains six cells across N ∈ {50, 200, 600} (~$10 total). Total project compute spend: **~$50**. Detailed methodology and per-cell tables in `research/results/reports/01–05`; architectural decisions and operational notes in `docs/adr/008–009`.

Conclusion

Hidden bias in LLM financial advisors is real, measurable, and large. It is induced equally well at the prompt level and the weight level, with effect sizes that survive judge-model independence and cross-model replication. Conventional plan-quality scoring fails to detect the directional shift in every model where it occurs. For deployment of LLM-based financial advice — particularly under commission-driven incentives — quality benchmarks alone are inadequate evidence of suitability; an explicit philosophy-axis audit, against a fixed-baseline reference, is needed.